

Federated Learning Results TP2

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1 Introduction

This report presents a comparison between three federated learning strategies: **FedAvg**, **FedProx**, and **SCAFFOLD**, using validation accuracy and training loss across multiple communication rounds. Each strategy was tested with different values of the α (data heterogeneity) parameter, and in the case of SCAFFOLD and FedProx, also with the μ parameter (proximal term coefficient).

2 Validation Accuracy by Strategy

2.1 FedAvg

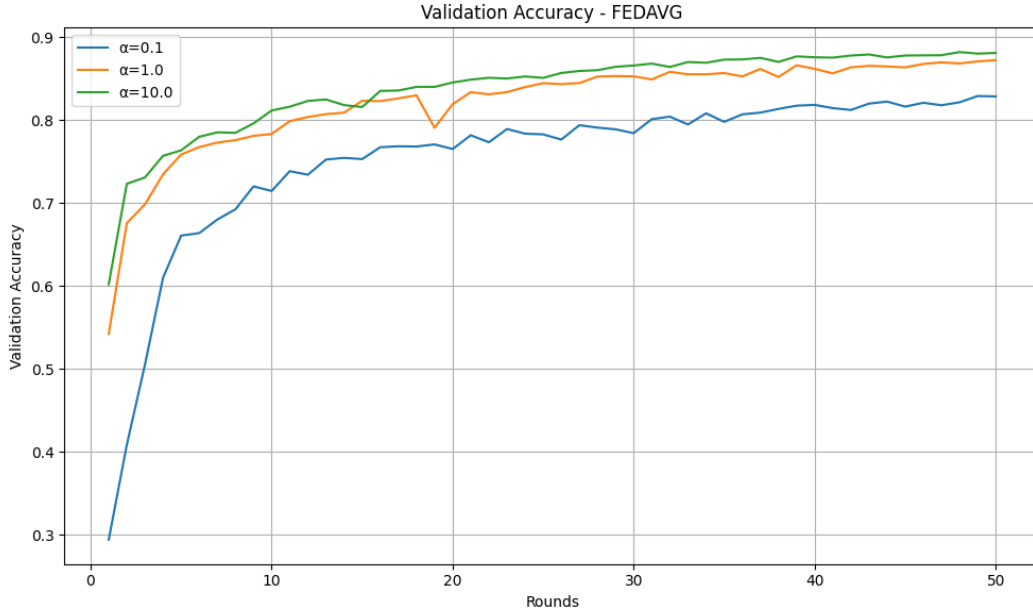


Figure 1: Validation Accuracy for FedAvg with different α values

2.2 FedProx

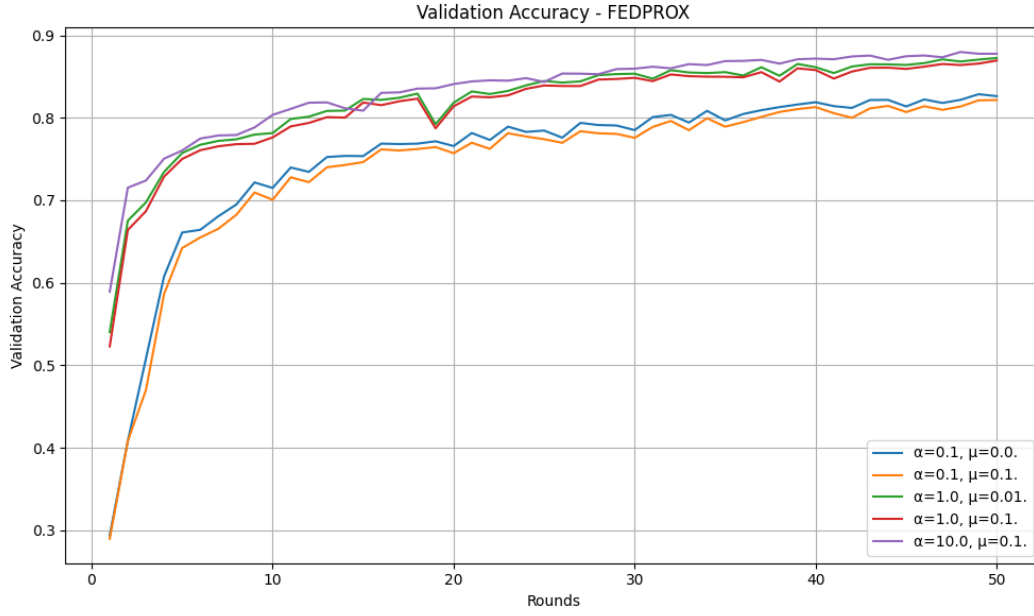


Figure 2: Validation Accuracy for FedProx with different α and μ values

2.3 SCAFFOLD

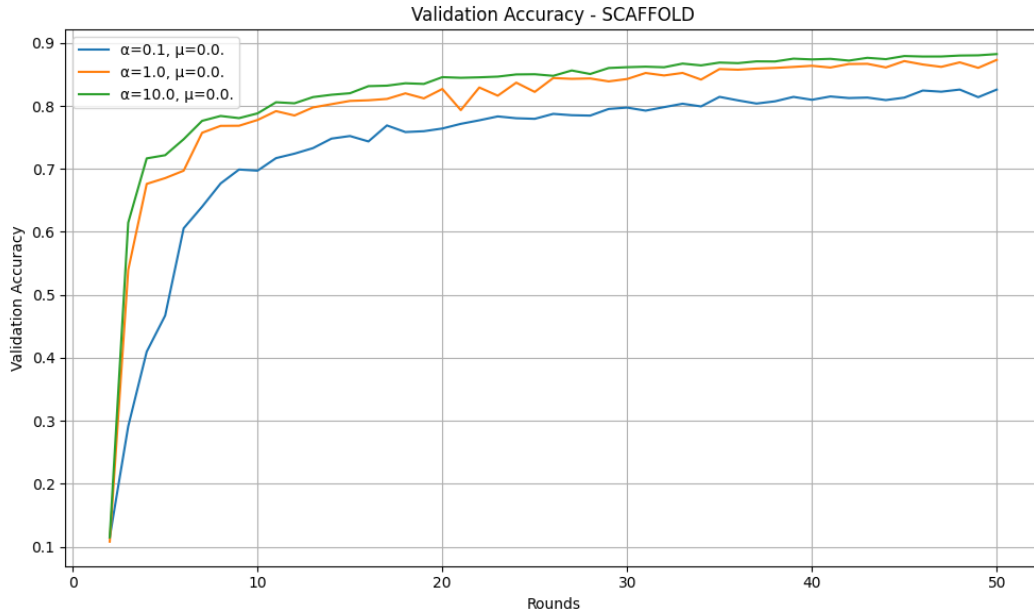


Figure 3: Validation Accuracy for SCAFFOLD with different α and μ values

2.4 Comparison of All Strategies

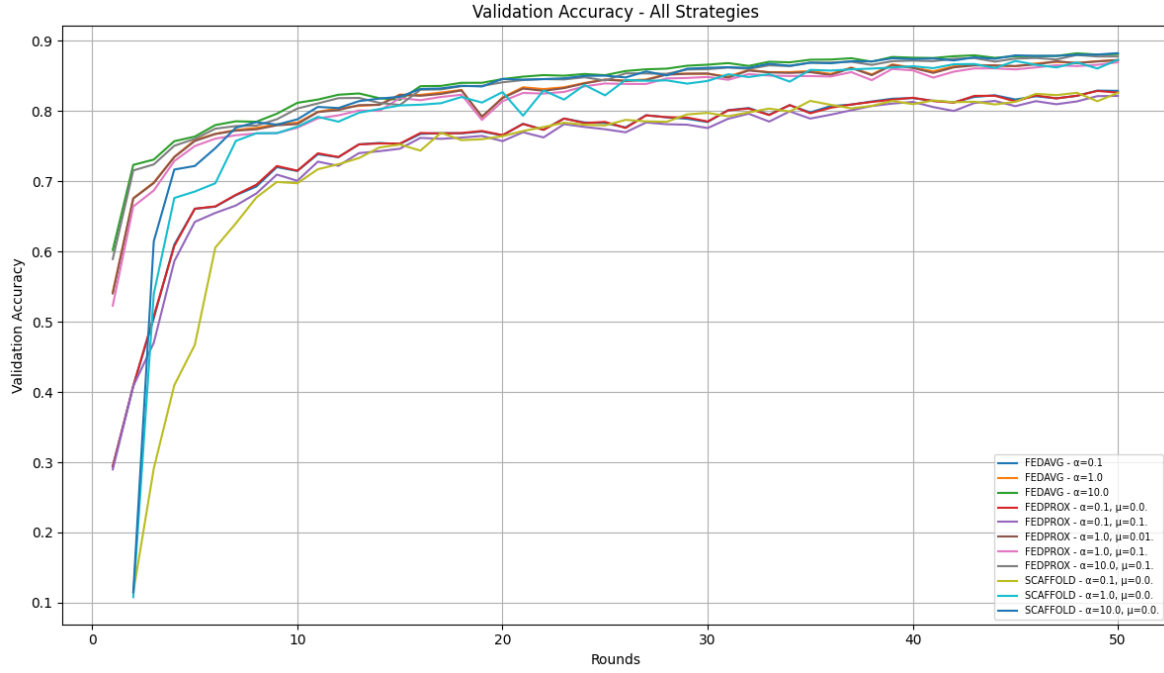


Figure 4: Comparison of Validation Accuracy for All Strategies

3 Training Loss by Strategy

3.1 FedAvg

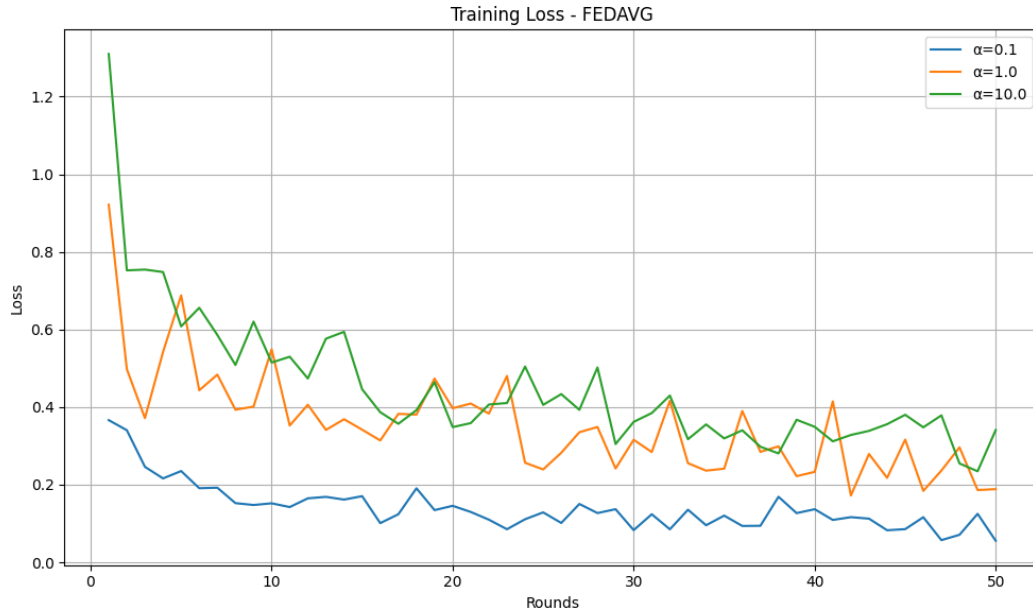


Figure 5: Training Loss for FedAvg

3.2 FedProx

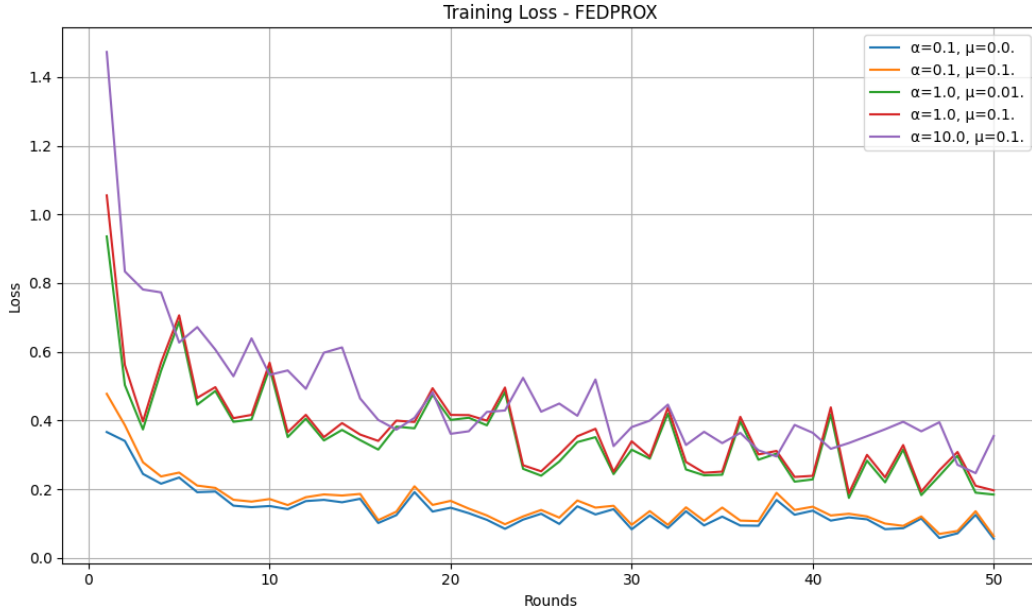


Figure 6: Training Loss for FedProx

3.3 SCAFFOLD

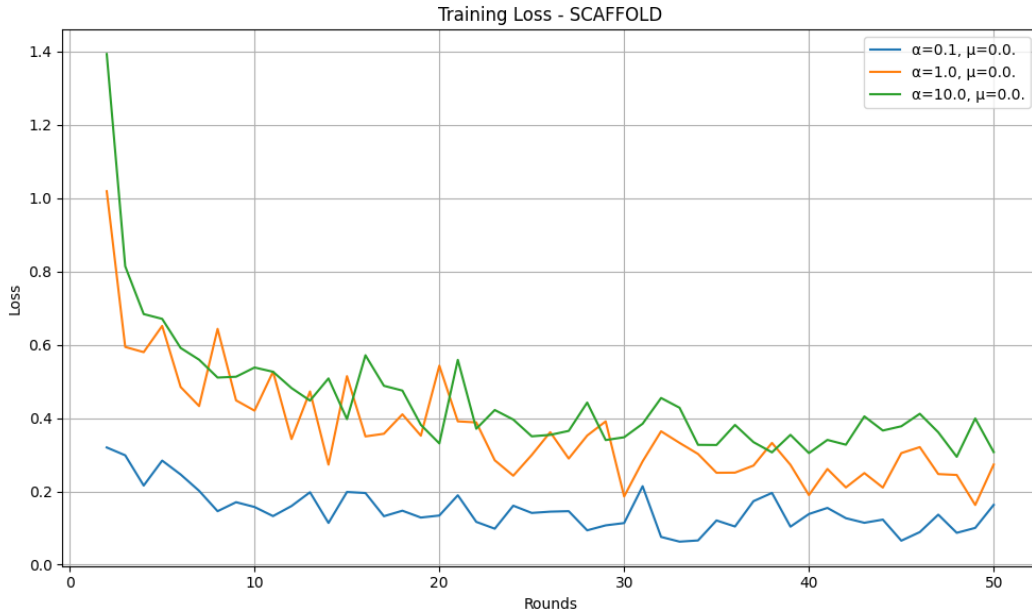


Figure 7: Training Loss for SCAFFOLD

3.4 Comparison of All Strategies

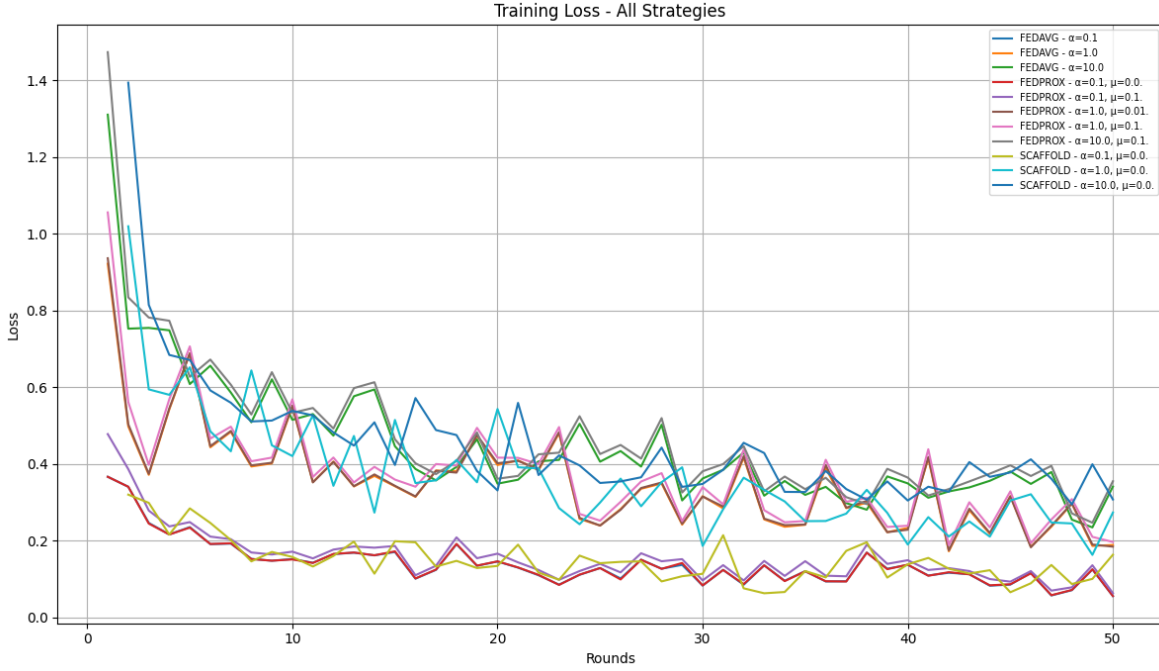


Figure 8: Comparison of Training Loss for All Strategies

4 Conclusion

From the results, we can observe that all three strategies converge effectively, with varying performance depending on α and μ values. SCAFFOLD and FedProx show improvements in certain settings, especially when μ is tuned properly. Validation accuracy and training loss plots help us understand convergence behavior and stability under non-iid conditions.